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COMMERCIAL ITEM DESCRIPTION

THREAD AND FORM ROLLING MACHINES, ROTARY DIE, POWER DRIVEN, HORIZONTAL

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers horizontal, rotary die, form- and thread-rolling machines intended for use in production rolling of threads and forms on tube and solid stock.

2. CLASSIFICATION. The thread and form rolling machines shall be of the following types, classes, styles, and sizes. The type, class, style, and size to be furnished shall be as specified (see 7.2(b)).

Type I - Two-die
Type II - Three-die

Class 1 - Manual
Class 2 - Automatic
Class 3 - Computer numerical control

Style A - Throughfeed and radial infeed
Style B - Throughfeed
Style C - Radial infeed

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3449

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Size 1 - 1/2-inch diameter
Size 2 - 1-inch diameter
Size 3 - 1 1/2-inch diameter
Size 4 - 2-inch diameter
Size 5 - 3-inch diameter
Size 6 - 4-inch diameter
Size 7 - 6-inch diameter

3. SALIENT CHARACTERISTICS

3.1 General requirements. The machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. The thread and form rolling machine shall include, as a minimum, the following components:

3.2.1 Base. The base shall support the other elements of the machine and shall incorporate materials, shape, and mass to minimize adverse vibration and deflection. The base shall provide a means for leveling and anchoring the machine.

3.2.2 Motor and power transmission. Unless otherwise specified (see 7.2(c)), the spindle motor horsepower shall be the manufacturer's standard for the machine. The system shall transmit power from the motor to the spindles without restricting spindle assembly movement within the diametrical capacity of the machine. The power transmission shall provide for spindle speed adjustment as specified (see 7.2(d)).

3.2.3 Spindle assembly. The spindle assembly shall withstand continuous and intermittent rolling forces developed during operation. Unless otherwise specified (see 7.2(e)), the rolling force shall be that considered by the manufacturer as standard for the machine. When specified (see 7.2(f)), the spindle rotation shall be reversible. Unless otherwise specified (see 7.2(g)), a means shall be provided to:

a. Adjust the spindle attitude
b. Adjust the spindle dies laterally
c. Adjust the spindles vertically, type I machine only
d. Adjust the center distance between the spindles

3.2.4 Workpiece support, type I. A workpiece support shall be provided that utilizes pads of different materials as specified (see 7.2(h)) and workpiece roller supports as specified (see 7.2(i)).

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3.2.5 Die set. The thread and form rolling die sets shall be supplied as specified (see 7.2(j)).

3.2.6 Controls.

3.2.6.1 Control panel. Unless otherwise specified (see 7.2(k)), class 1 and 2 thread and form rolling machines shall have, as a minimum, a control panel with the following manual controls and switches:

a. System keyed lock
b. Emergency stop button
c. Power on/off switches
d. Spindle infeed, for style A and C machines only
e. Coolant flow
f. Spindle rotation

3.2.6.2 Automatic control cycle. In addition to manual controls, class 2 thread and form rolling machines shall have an automatic control cycle. Unless otherwise specified (see 7.2(l)), the automatic control cycle, as a minimum, shall have the following control functions:

a. Spindle position
b. Die rotation
c. Spindle rotation
d. Coolant flow
e. Pause, for workpiece movement and placement

3.2.7 Computer numerical control. The class 3 thread and form rolling machine shall be supplied with a computer numerical control (CNC) system. The CNC system shall be either a personal computer (PC) open-architecture type or other software programmable microprocessor design. The CNC system shall be industrial grade in accordance with accepted industry practices and shall meet Government requirements as specified (see 7.2(m)).

3.2.7.1 Hardware features. Unless otherwise specified (see 7.2(n)), the CNC system shall include, as a minimum, the following hardware:

a. One or more 32-bit microprocessors with math coprocessor
b. An American National Standards Institute/Telecommunications Industry Association/Electronic Industries Association (ANSI/TIA/EIA)-232, "Interface Between Data Terminal Equipment and Data Circuit-Termination Equipment Employing Serial Binary Data Interchange", communications port
c. 6-inch monochrome monitor
d. Manual data input (MDI) device: PC keyboard or alphanumeric keyboard with dedicated keys for specific machine functions

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3.2.7.2 Control panel. Unless otherwise specified (see 7.2(o)), the CNC system shall include, as a minimum, a control panel with the following manual controls and switches:

a. System keyed lock
b. Emergency stop button
c. Power on/off switches
d. CNC system start/stop switches (cycle start/stop switches)
e. Spindle motor power and spindle speed display

3.2.7.3 Operating modes. Unless otherwise specified (see 7.2(p)), the CNC system shall include, as a minimum, the following operating modes:

a. Manual operation mode
b. MDI mode
c. Conversational mode
d. Automatic mode
e. Machine diagnostic/maintenance/monitoring modes

3.2.7.4 Software features. Unless otherwise specified (see 7.2(q)), the CNC system shall include, as a minimum, the following software programming and control features:

a. Control of spindle axes movement and rotation
b. Inch/metric switchable programming
c. Decimal point programming
d. Reference zero
e. Standard EIA-494, "32 Bit Binary CL (BCL) and 7 Bit ASCII CL (ACL) Exchange Input Format for Numerically Controlled Machines", codes compatibility

3.2.8 Lubrication system. Unless otherwise specified (see 7.2(r)), the thread and form rolling machine shall be equipped with an automatic lubricating system to direct oil flow from the reservoir to all machine components requiring continuous lubrication. A means shall be provided for determining oil level in the reservoir. The automatic lubricating system shall include a cleanable or replaceable filter. Upon loss of pressure in the automatic lubrication system, the thread and form rolling machine shall automatically stop. Machine parts requiring periodic manual lubrication shall be readily accessible through doors or removable panels.

3.2.9 Coolant system. Unless otherwise specified (see 7.2(s)), the rolling dies shall be flood cooled. As a minimum, the coolant system shall consist of a reservoir, pump, strainer, and all necessary piping. The system shall be equipped with a method to control the rate of coolant flow from the reservoir and to direct coolant flow to the die rolling area. When specified (see 7.2(t)), an optional filtration system shall be provided.

3.2.10 Hydraulic system. If a hydraulic system is provided, it shall conform to the requirements of National Fluid Power Association (NFPA) T2.24.1, "Hydraulic Fluid Power - System Standard For Stationary Industrial Machinery First Edition". The hydraulic system shall be complete and include all pumps, valves, piping, cylinders, fluid level indicator, and controls.

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Means shall be provided to prevent damage to the components from overpressure. A filter system shall be provided to ensure delivery of clean fluid.

3.2.11 Electrical system. Unless otherwise specified (see 7.2(u)), the electrical system shall conform to the National Fire Prevention Association (ANSI/NFPA) 79, "Electrical Standard for Industrial Machinery". The machine shall have dual 230/460-volt, 3-phase, 60-hertz circuit capability and shall be wired for the voltage specified (see 7.2(v)). Unless otherwise specified (see 7.2(w)), the control system power shall be provided by a separate 120-volt, single-phase, 60-hertz circuit from the machine main power supply.

3.2.12 Drive motors. The spindle, hydraulic (when provided), and coolant drive motors shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG-1, "Motors and Generators". Unless otherwise specified (see 7.2(x)), the motors shall have open dripproof enclosures. Motor controls shall conform to NEMA ICS-1, "Industrial Control and Systems General Requirements", and NEMA ICS-3, "Industrial Control and Systems Factory Built Assemblies". The spindle drive motor shall be capable of continuous thread-rolling operation at the maximum thread size specified for the machine.

3.2.13 Optional accessories. Optional accessories shall be provided as specified (see 7.2(y)).

3.3 Performance. Unless otherwise specified (see 7.2(z)), the thread and form rolling machine shall meet the alignment and performance requirements shown in tables I and II, respectively, when tested in accordance with sections 5.3 through 5.3.5.

TABLE I. Alignment requirements.

Parameter	Performance
Spindle runout	0.002 inch TIR[1]
Spindle endplay	0.001 inch TIR[1]
Spindle axial alignment	0.002 inch TIR[1]

[1] TIR is the total indicated reading.

TABLE II. Performance requirements.

Machine size	ASME B1.1 thread size
1	1/4-20UNC-3A
2	1/2-13UNC-3A
3	3/4-10UNC-3A
4	1-8UNC-3A
5	1 1/2-6UNC-3A
6	2-4 1/2UNC-3A
7	3-4UNC-3A

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3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(aa)).

3.5 Safety and health requirements. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910, "Occupational Safety and Health Standards for General Industry".

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(bb)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(cc)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(dd)), the nameplate shall contain the following information:

a. Nomenclature
b. Manufacturer's name
c. Serial number
d. Machine model designation
e. Power input (volts, total amperes, phase, frequency)
d. Short-circuit/over-current rating
f. Contract number or order number
g. National stock number
h. Date of manufacture

3.8 Lubrication plate or chart. When specified (see 7.2(ee)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application
b. Servicing interval
c. Type of lubricant(s) with Society of Automotive Engineers (SAE) number or lubricant identifier

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user’s health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

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5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Inspection. The thread and form rolling machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(ff)), the thread and form rolling machine shall be tested in accordance with sections 5.3.1 through 5.3.5.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(gg)), the manufacturer shall be responsible for supplying all tooling and material(s) and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(hh)). Unless otherwise specified (see 7.2(ii)), the machine shall be tested in an indoor facility with the temperature between 40 and 110° F and the relative humidity between 20 and 95 percent, noncondensing.

5.3.2 Operational test. The thread and form rolling machine shall be operated in accordance with the manufacturer's standard operational test procedures. The thread-rolling machine shall be operated at no load for not less than 4 hours. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Alignment tests. The thread and form rolling machine alignment tests shall be conducted before and after the performance tests and shall meet the criteria shown in table I. Changes in alignment shall be cause for rejection.

5.3.4 Performance tests. The performance tests shall be conducted in accordance with sections 5.3.4.1 to 5.3.4.2. There shall be no evidence of overloading, overheating, excessive vibration, or deformation of any part of the machine during or after the test.

5.3.4.1 Test workpiece. The test workpiece material shall be S41400 stainless steel. The initial test workpiece diameter shall be the manufacturer's suggested diameter to produce the thread size shown in table II. The length of the radial infeed test workpiece shall be 1 1/2 inches long with a shoulder 1/4 inch from the end. The throughfeed test workpiece shall be 12 inches long.

5.3.4.2 Rolling method. Style A and C machines shall thread the five 12-inch long test workpieces using the throughfeed method and thread the 5 test workpieces with a shoulder using the radial infeed method. Style A and B machines shall thread the five 12-inch long test workpieces using the throughfeed method. The completed test workpieces shall meet the requirements shown in table II and American Society of Mechanical Engineers (ASME) B1.1, "Unified Inch Screw Threads (UN and UNR Thread Form)".

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5.3.5 Optional tests. Optional tests shall be conducted as specified (see 7.2(jj)).

5.4 Acceptance. Unless otherwise specified (see 7.2(kk)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the alignment and performance tests shall be compared with the requirements shown in tables I and II, respectively. Failure of the machine to meet the requirements shown in tables I and II shall be cause for rejection.

6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the machine shall be preserved, packed, and marked as specified in the acquisition order (see 7.2(ll)).

7. NOTES

7.1 Sources of referenced documents.

7.1.1 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

7.1.2 ANSI standards. Copies of ANSI standards may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

7.1.3 ANSI/NFPA standards. Copies of ANSI/NFPA standards may be obtained from the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269-9101.

7.1.4 ASME standards. Copies of ASME standards may be obtained from the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

7.1.5 NEMA standards. Copies of NEMA standards may be obtained from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

7.1.6 NFPA standards. Copies of ANSI/NFPA standards may be obtained from the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269-9101.

7.1.7 NFPA standards. Copies of NFPA standards may be obtained from the National Fluid Power Association, 3333 North Mayfair Road, Suite 3111, Milwaukee, WI 53222-3219.

7.1.8 OSHA standards. Copies of OSHA standards may be obtained from U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

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7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document
b. Type, class, style, and size of thread-rolling machine required (see 2)
c. Spindle motor horsepower, if different (see 3.2.2)
d. Spindle speed adjustment, as specified (see 3.2.2)
e. Rolling force, if different (see 3.2.3)
f. Reversing spindle rotation, if required (see 3.2.3)
g. Spindle adjustment, if different (see 3.2.3)
h. Workpiece support pad material, as specified (see 3.2.4)
i. Workpiece roller support, as specified (see 3.2.4)
j. Die sets, as specified (see 3.2.5)
k. Manual controls, if different (see 3.2.6.1)
l. Automatic control cycle, if different (see 3.2.6.2)
m. CNC system requirements, as specified (see 3.2.7)
n. CNC hardware features, if different (see 3.2.7.1)
o. CNC control panel, if different (see 3.2.7.2)
p. CNC operating modes, if different (see 3.2.7.3)
q. CNC software features, if different (see 3.2.7.4)
r. Lubrication system, if different (see 3.2.8)
s. Coolant system, if different (see 3.2.9)
t. Coolant filtration system, if required (see 3.2.9)
u. Electrical system industrial standard, if different (see 3.2.11)
v. Electrical system voltage, as specified (see 3.2.11)
w. Control system voltage, if different (see 3.2.11)
x. Motor enclosure, if different (see 3.2.12)
y. Optional accessories, as specified (see 3.2.13)
z. Performance, if different (see 3.3)
aa. Machine dimensions and weight, as specified (see 3.4)
bb. System of units required (see 3.6)
cc. Dual system of units, if required (see 3.6)
dd. Nameplate, if different (see 3.7)
ee. Lubrication plate or chart, if required (see 3.8)
ff. Testing, if different (see 5.3)
gg. Test responsibility, if different (see 5.3.1)
hh. Certification of calibration, as specified (see 5.3.1)
ii. Test environmental conditions, if different (see 5.3.1)
jj. Optional tests, as specified (see 5.3.5)
kk. Acceptance, if different (see 5.4)
ll. Packaging requirements, if different (see 6.1)

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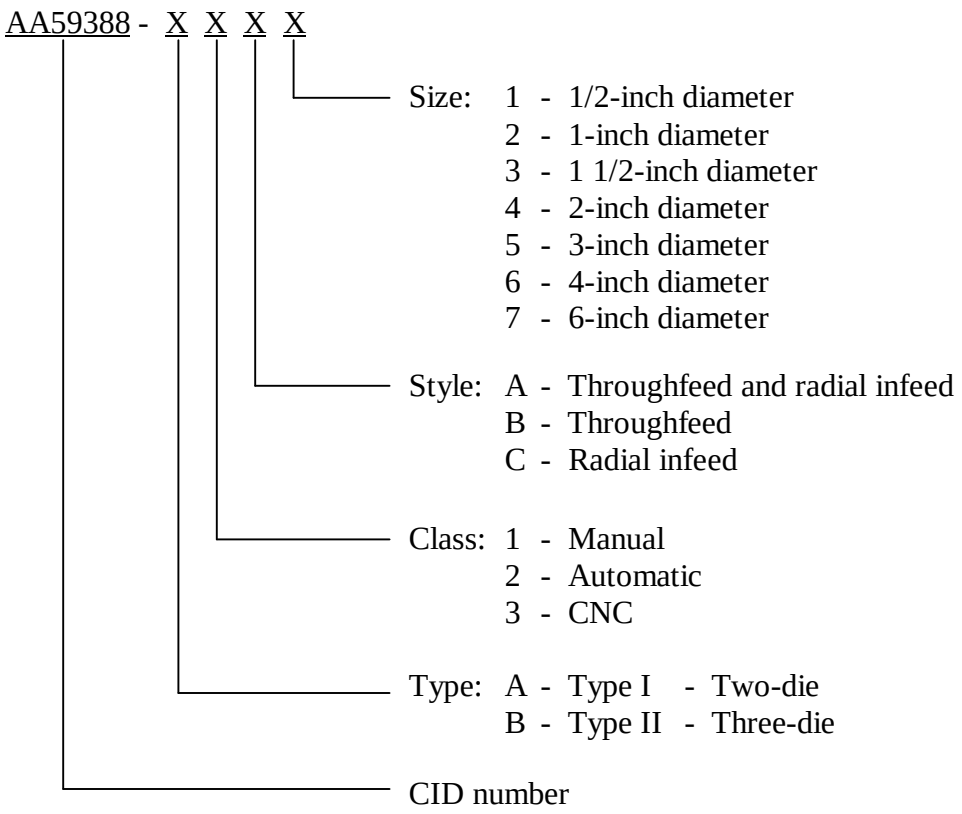
7.3 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.



MILITARY INTERESTS:

Custodians:
Army - AL
Navy - SH
Air Force - 99

Reviewers:
Navy - MC
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 6FET

Preparing activity:
DLA - GS

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